

# SOLUTIONS

*ISSUE 1 - APRIL 2024*

$$\begin{aligned}
 & \frac{x-6}{2022} + \frac{x-5}{2023} + \frac{x-4}{2024} = 3 \\
 \Rightarrow & \frac{x-6}{2022} + \frac{x-5}{2023} + \frac{x-4}{2024} - 3 = 0 \\
 \Rightarrow & \frac{x-6}{2022} - 1 + \frac{x-5}{2023} - 1 + \frac{x-4}{2024} - 1 = 0 \\
 \Rightarrow & \frac{x-6}{2022} - \frac{2022}{2022} + \frac{x-5}{2023} - \frac{2023}{2023} + \frac{x-4}{2024} - \frac{2024}{2024} = 0 \\
 \Rightarrow & \frac{x-6-2022}{2022} + \frac{x-5-2023}{2023} + \frac{x-4-2024}{2024} = 0 \\
 \Rightarrow & \frac{x-2028}{2022} + \frac{x-2028}{2023} + \frac{x-2028}{2024} = 0 \\
 & \therefore x-2028 = 0 \\
 & \wedge x = 2028
 \end{aligned}$$

*ISSUE 2 - MAY 2024*

$$\begin{aligned}
 2 &= \sqrt{x\sqrt{x\sqrt{x\sqrt{x\cdots}}}} \\
 \Rightarrow 2^2 &= x\sqrt{x\sqrt{x\sqrt{x\cdots}}} \\
 \Rightarrow 4 &= x\sqrt{x\sqrt{x\sqrt{x\cdots}}} \\
 \Rightarrow \frac{4}{x} &= \sqrt{x\sqrt{x\sqrt{x\sqrt{x\cdots}}}} \\
 &\therefore \frac{4}{x} = 2 \\
 \Rightarrow 4 &= 2x \\
 &\wedge x = 2
 \end{aligned}$$

Cont. Overl.

*ISSUE 3 - JUNE 2024*

$$\begin{aligned}
\frac{9^{-1}}{81^x} &= 9^{x^2} \\
\Rightarrow \frac{1}{81^x \cdot 9} &= 9^{x^2} \\
\Rightarrow 1 &= 9^{x^2} \cdot 81^x \cdot 9 \\
\Rightarrow 1 &= 9^{x^2} \cdot 9^{2x} \cdot 9 \\
\Rightarrow 1 &= 9^{x^2+2x+1} \\
\therefore \log_9 1 &= \log_9 9^{x^2+2x+1} \\
\Rightarrow 0 &= x^2 + 2x + 1 \\
\Rightarrow 0 &= (x+1)^2 \\
\wedge x &= -1
\end{aligned}$$

*ISSUE 4 - JULY/AUGUST 2024*

$$\begin{aligned}
2 \log_2 x - \frac{1}{2} \log_2 x &= 3 \\
\Rightarrow \log_2 x^2 - \log_2 x^{\frac{1}{2}} &= 3 \\
\Rightarrow \log_2 \frac{x^2}{x^{\frac{1}{2}}} &= 3 \\
\Rightarrow \log_2 x^{\frac{3}{2}} &= 3 \\
\therefore x^{\frac{3}{2}} &= 8 \\
\Rightarrow x^{\frac{1}{2}} &= 2 \\
\wedge x &= 4
\end{aligned}$$

*ISSUE 5 - SEPTEMBER 2024*

$$\begin{aligned}
\frac{9^{2x+1} \times 3^{4-3x}}{27^{2-x}} &= 9 \\
\Rightarrow \frac{3^{2(2x+1)} \times 3^{4-3x}}{3^{3(2-x)}} &= 9 \\
\Rightarrow 3^{2(2x+1)+(4-3x)-3(2-x)} &= 9 \\
\Rightarrow 3^{(4x+2)+(4-3x)-(6-3x)} &= 9 \\
\therefore 3^{4x} &= 9 \\
\wedge x &= \frac{1}{2}
\end{aligned}$$